ព្រះរាជាណាចក្រកម្ពុជា

ជាតិ សាសនា ព្រះមហាក្សត្រ

Software Engineering

Project title: “ Library Borrowing and Book Tracking System”

Final Report of the project

Group 4-I4GIC-A

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Academic Year 2025-2026

Tables Content

Library and borrow book tracking system

I. Introduction

1.1. Project Overview

The Library and Borrow Book Tracking System is a computerized system designed to manage library resources and monitor book borrowing activities efficiently. This system replaces traditional manual record-keeping with a digital solution that allows librarians and users to manage books, members, and borrowing records accurately. By using a structured database, the system ensures that all library operations such as adding books, tracking borrowed items, and managing returns are organized and easy to maintain.

1.2. Purpose of the Library Management System

The main purpose of the Library Management System is to simplify and automate the process of managing library activities. The system helps librarians keep track of available books, borrowed books, and borrower information in real time. It reduces human errors, saves time, and improves the overall efficiency of the library. Additionally, it allows users to easily search for books, view borrowing history, and understand due dates, making library services more accessible and user-friendly.

1.3 Scope of the project

The scope of this project covers essential library operations including book management, user management, and borrow/return tracking. The system supports features such as adding and updating book information, registering library members, recording book borrowing and return dates, and generating reports on library usage. However, advanced features such as online payments or multi-branch library management are beyond the scope of this project. The system is intended for small to medium-sized libraries such as school or college libraries.

II. Project Planning

2.1. Development Methodology

The development of the **Library Borrowing and Book Tracking System** followed a **modular and incremental approach**.

The system was divided into several modules, such as:

- User Management
- Book Management
- Borrowing and Returning System
- Database Management

Each module was developed and tested separately before being integrated into the full system. This approach made it easier to find errors and improve system quality.

The development process included the following steps:

1. Analyze System requirements
2. Design the System architecture and database
3. Develop restful APIs using Spring Boot
4. Create database tables using MYSQL
5. Connect backend APIs with the database
6. Test each function as login, book borrowing, and returning
7. Integrate all modules into one complete system

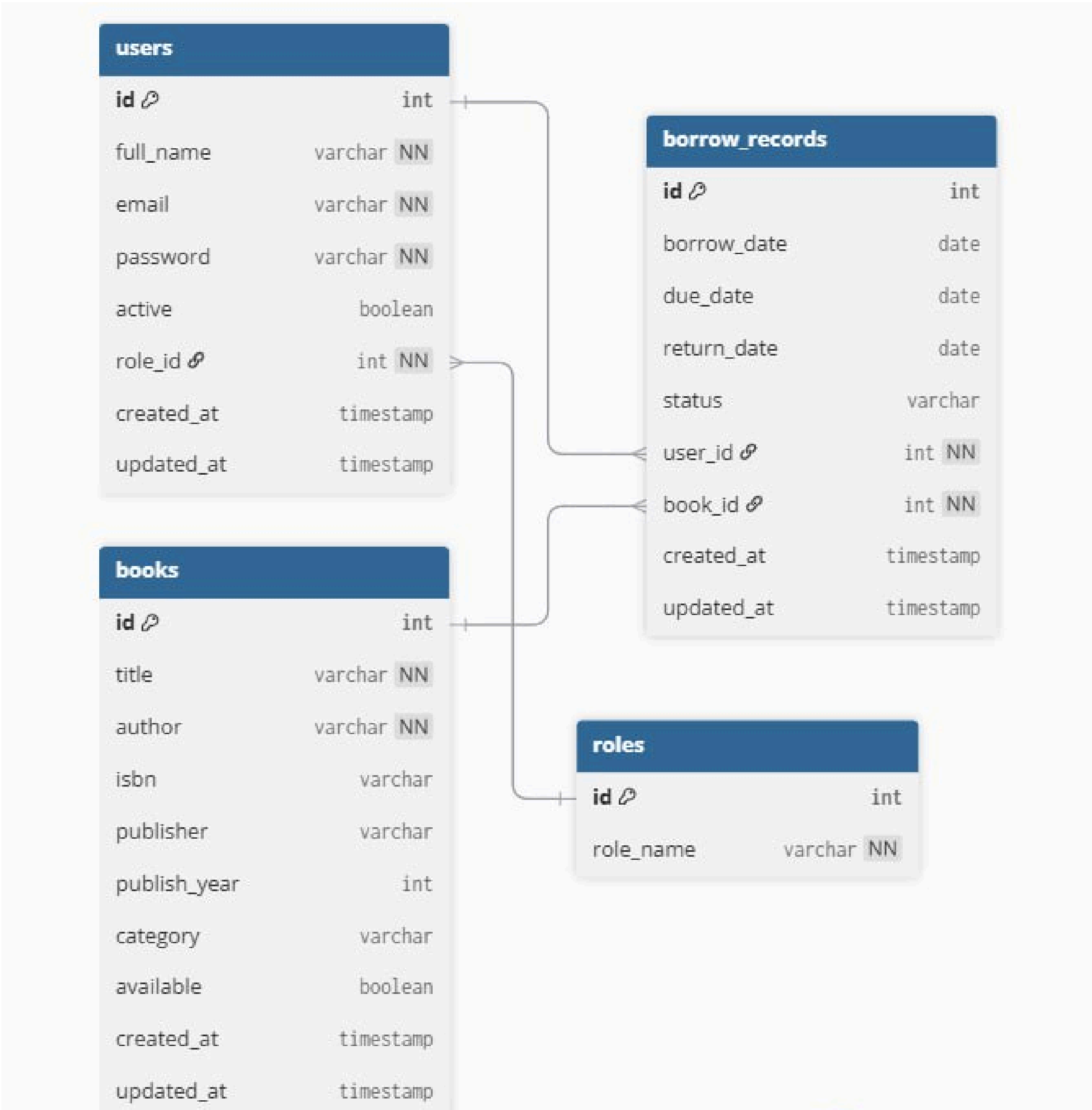
This methodology helped ensure that the system was well organized, reliable, and easy to maintain.

III. Task Division

3.1 Team Roles and Responsibilities**Task Division**

Name	**Role**	**Responsibilities**
NETH Tithyarita	Leader	Security
MENG Leangei	Member	Crud (secondary)
KAING Sophea	Member	Crud (main)
DOEUNG Sreypy	Member	Database
GUOV Houysang	Member	Front-end

IV. Relational database design



4.1 Relational Database

- The system uses a relational database model
- Data is stored in separate tables to reduce redundancy
- Tables are connected using primary and foreign keys
- Designed following Third Normal Form (3NF)

4.2 Relationship

- Users ↔ Roles:
  One role can be assigned to many users (One-to-Many)
- Users ↔ Borrow_Records:
  One user can have multiple borrow records
- Books ↔ Borrow_Records:
  One book can appear in many borrow records
- Borrow_Records acts as a linking table between users and books

4.3 Table-Specific constraints

Constraints are rules applied to table columns to maintain data accuracy and integrity.

A. USERS Table Constraints

Column	Type	Constraint
Id	int	PK, AUTO_INCREMENT
Full-name	varchar	NOT NULL
email	varchar	UNIQUE, NOT NULL
password	varchar	NOT NULL
active	boolean	NULLABLE
role_id	int	FK → roles.id, NOT NULL
create_at	timestamp	DEFAULT CURRENT_TIMESTAMP
update_at	timestamp	DEFAULT CURRENT_TIMESTAMP

B. ROLES Table Constraints

Column	Type	Constraint
Id	Int	PK, AUTO_INCREMENT
Role-name	Varchar	Unique, not null

C. BOOK Table Constraints

Column	Type	Constraint
id	int	PK, Auto Increment
title	varchar	NOT NULL
author	varchar	NOT NULL
isbn	varchar	UNIQUE
publisher	varchar	NULLABLE
publisher_year	int	NULLABLE
category	varchar	NULLABLE
available	boolean	NULLABLE
create_at	timestamp	DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp	DEFAULT CURRENT_TIMESTAMP

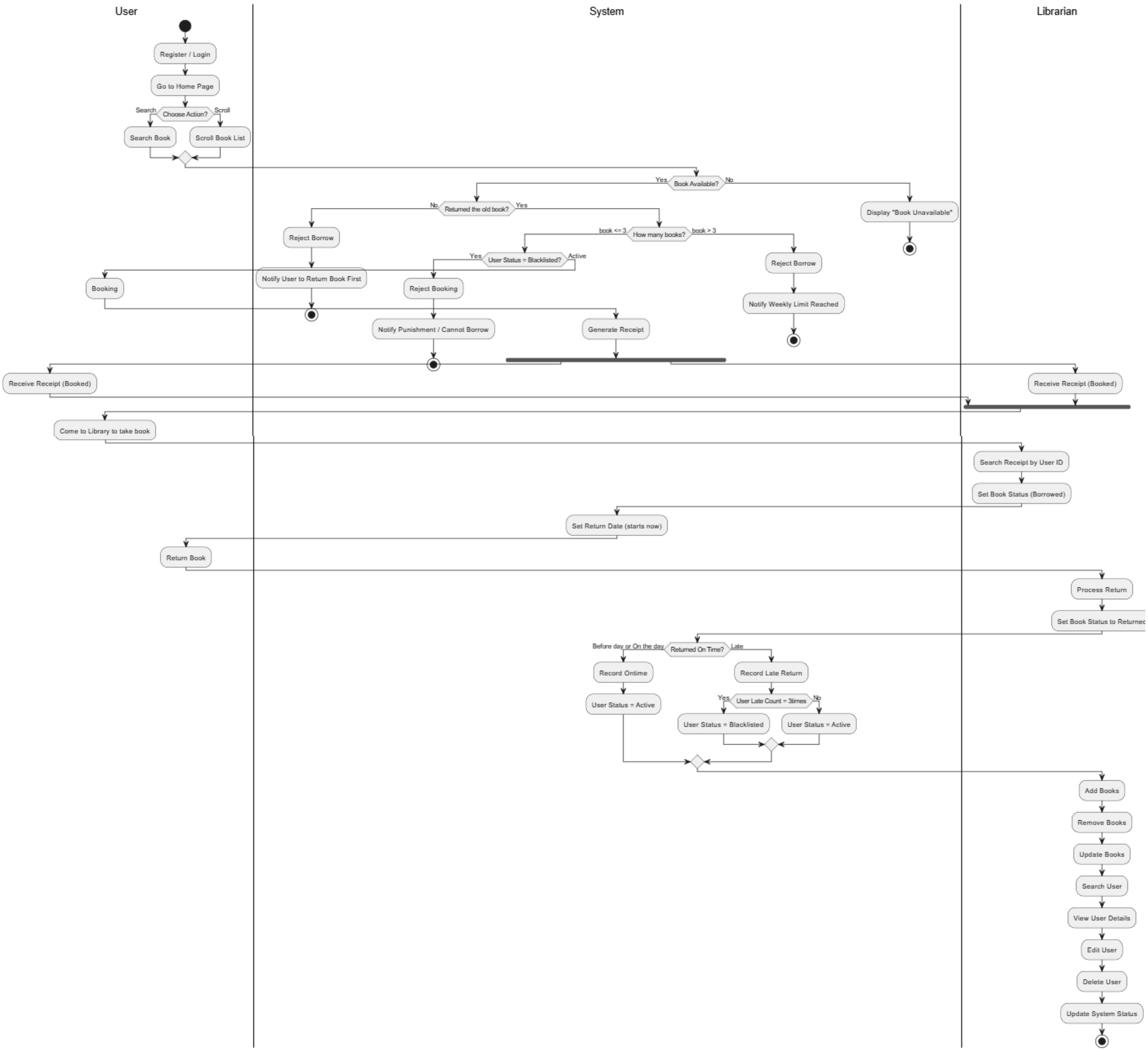
D. BORROWS Table Constraints

Column	Type	Constraint
id	int	PK, AUTO_INCREMENT
borrow_date	date	NULLABLE
due_date	date	NULLABLE
return_date	date	NULLABLE
status	varchar	NULLABLE
user_id	int	FK → users.id, NOT NULL
book_id	int	FK → books.id, NOT NULL
created_id	timestamp	DEFAULT CURRENT_TIMESTAMP
updated_id	timestamp	DEFAULT CURRENT_TIMESTAMP

Referential Integrity

- Foreign key constraints prevent invalid references
- Deleting a user or book is restricted if borrow records exist
- Ensures consistent and reliable data

V. Draw PUML



User
System
Librarian
Register / Login
Go to Home Page
Search
Choose Action?
Scroll
Search Book
Scroll Book List
Yes
Book Available?
No
No
Returned the old book?
Yes
Display "Book Unavailable"
Reject Borrow
book <= 3
How many books?
book > 3
Yes
User Status = Blacklisted?
Active
Reject Borrow
Booking
Notify User to Return Book First
Reject Booking
Notify Weekly Limit Reached
Notify Punishment / Cannot Borrow
Generate Receipt
Receive Receipt (Booked)
Receive Receipt (Booked)
Come to Library to take book
Search Receipt by User ID
Set Book Status (Borrowed)
Set Return Date (starts now)
Return Book
Process Return
Set Book Status to Returned
Before day or On the day
Returned On Time?
Late
Record Ontime
Record Late Return
User Status = Active
Yes
User Late Count = 3times
No
User Status = Blacklisted
User Status = Active
Add Books
Remove Books
Update Books
Search User
View User Details
Edit User
Delete User
Update System Status

Description of UML

- User Registration and Login :
    - The process begins when a user registers or logs into the system. After they login successful, the user is redirected to the home page.Borrowing Conditions Check:
        - Before allowing a new borrowing request, the system verifies:
            - Whether the user has returned previously borrowed books
                - If not returned, the system rejects the borrow request and notifies the user to return the old book first.
            - Whether the user has reached the weekly borrowing limit (maximum 3 books)
                - If the request more than 3, the request is rejected and the user is notified
- Browsing and Searching for Books :
    - On the home page, the user can search for a specific book, or scroll through the list of available books.
    - The system then checks whether the selected book is available.
        - If the book is not available, the system displays a *"Book Unavailable"* message so the user can't borrow it.
        - If the book is available, the system continues to the borrowing conditions.
- User Status Verification:
    - The system checks the user status:
        - If the user is blacklisted, the system rejects the booking and notifies the user of the punishment.
        - If the user is active, the booking process continues.
- Booking Process:
    - The user proceeds to book the book.
        - The system generates a booking record.
        - The record is sent to both the user and the librarian simultaneously.
- Borrowing the Book
    - After booking:
        - The user comes to the library to take the book.
        - The librarian searches for the booking using the User ID and sets the book status to "Borrowed".
        - The system automatically sets the return date, starting from the borrowing day.
- Book Return Process
    - When the user returns the book:
        - The librarian processes the return and updates the book status to "Returned"
    - The system checks the return timing:
        - On-time return : User status remains Active
        - Late return (3 times): the user is blacklisted
- Librarian Management Activities
    - The librarian can also perform administrative tasks such as:

- Adding, removing, and updating book
- Searching for users, viewing, editing, or deleting user details
- Updating system status,(borrowed & return)

VI. Design Workflow

6.1. System Architecture

The Library and Borrow Book Tracking System follows a client–server architecture. The system consists of three main components: the client interface, the application server, and the database. Users interact with the system through a web-based interface, while the server handles business logic and data processing. The database stores all information related to books, users, and borrowing transactions.

This architecture ensures separation of concerns, improves system scalability, and allows easy maintenance and future enhancements.

Main Components:

- Client (Frontend): User interface for librarians and users
- Application Server (Backend): Handles requests, logic, and validation
- Database (MySQL): Stores books, users, roles and borrowing records

6.2. Layered Architecture(Controller, Service, Repository)

The system is designed using a three-layered architecture to improve code organization and maintainability.

A. Controller

The Controller layer handles incoming API requests from the client. It receives user input, performs basic validation, and forwards requests to the Service layer. It also returns responses back to the client.

Responsibilities:

- Handle HTTP requests (GET, POST, PUT, DELETE)
- Validate request data
- Send responses to the client

B. Service

The Service layer contains the core business logic of the system. It processes data received from the Controller layer and applies rules such as checking book availability or validating borrow limits.

Responsibilities:

- Apply business rules
- Process borrow and return logic
- Coordinate data between Controller and Repository layers

C. Repository

The Repository layer interacts directly with the database. It performs CRUD (Create, Read, Update, Delete) operations and abstracts database access from the Service layer.

Responsibilities:

- Execute database queries
- Manage data persistence
- Return data objects to the Service layer

6.3. Api endpoint

The system exposes RESTful API endpoints to allow communication between the client and server.

A. Book Management APIs

Base Path: /books

GET /books – View all books

GET /books/add – Show add book form

POST /books/save – Save new book

GET /books/edit/{id} – Show edit book form

POST /books/edit/{id} – Update book

POST /books/delete/{id} – Delete book

B. User Management APIs

- **General**

  GET /index – Welcome page

  GET /user/home – User home page

- **Search & Browse**

  GET /user/search – Search books

- **Borrow / Booking**

  GET /user/booking/{bookId} – Reserve a book by selecting it

  GET/user/borrowRecord/{bookId} - Select book

- **Account Management**

  GET /user/account – View account

  GET /user/account/edit – Edit profile

  POST /user/account/edit – Update profile

  GET /user/change_password – Change password page

  POST /user/change_password – Update password

  POST /user/account/change-password – Update password (alias)

C. Borrow Management APIs

- **BorrowRecord**

  GET /user/borrowRecord – View all borrow records

  POST /user/borrowRecord/add – Create borrow booking

  POST /user/borrowRecord/{id}/delete – Delete borrow record

D. Librarian Management APIs

Base Path: /librarian

- **Dashboard**

  GET /librarian/dashboard – Librarian dashboard

- **User Management**

  GET /librarian/manage_user – View/search users

  GET /librarian/add-user – Add user form

  POST /librarian/add-user – Save new user

  GET /librarian/edit-user/{id} – Edit user form

  POST /librarian/edit-user/{id} – Update user

  POST /librarian/manage_user/deactivate/{id} – Deactivate user

  POST /librarian/manage_user/reactivate/{id} – Reactivate user

  GET /librarian/user/{id}/history – View user borrow history

- **Borrow Management**

  GET /librarian/records – View pending borrows

  POST /librarian/borrow/confirm/{id} – Confirm borrow

  GET /librarian/process_return – Process returns

  POST /librarian/borrow/return/{id} – Confirm return

  POST /librarian/borrow/delete/{id} – Delete borrow record

  GET /librarian/records/search – Search borrow records

- **Redirect Helpers**

  GET /librarian/books → /books

  GET /librarian/manage-borrows → /librarian/records

E. Authentication APIs

Base Path: /api/auth

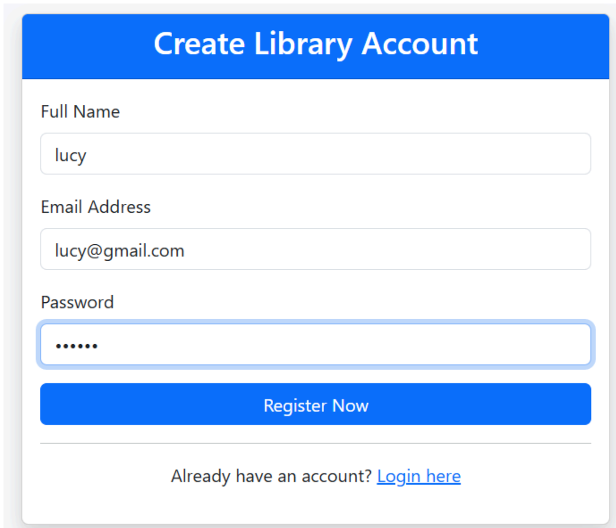
GET /api/auth/register – Show registration page

- POST /api/auth/register – Register new user

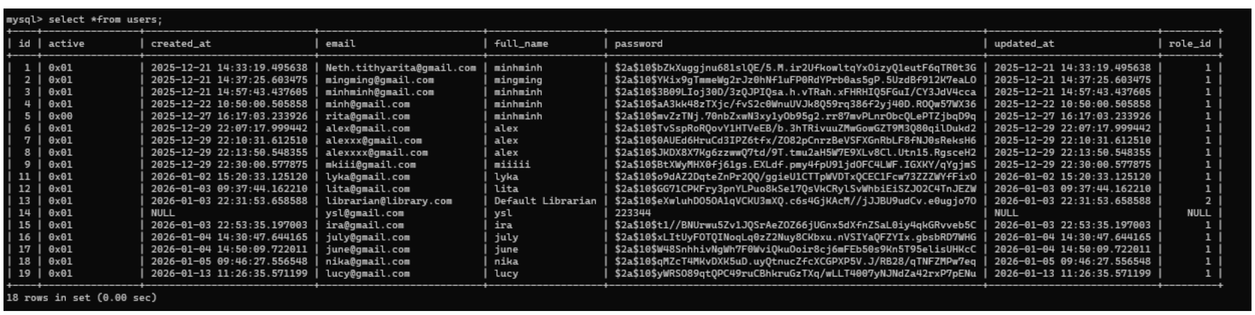
VII. Implementation

7.1. Login & User Management

- Here's Register Page
    - ★ If the student doesn't have the account yet, students need to register first.



- ★ After that the name will store in database(MYSQL)

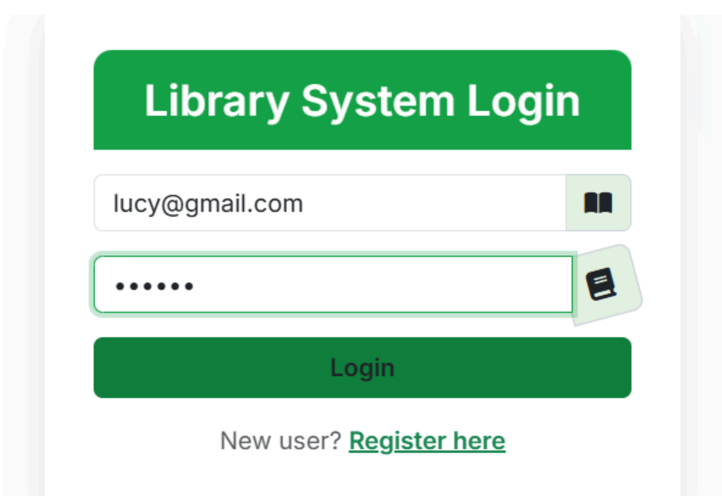


mysql> select *from users;

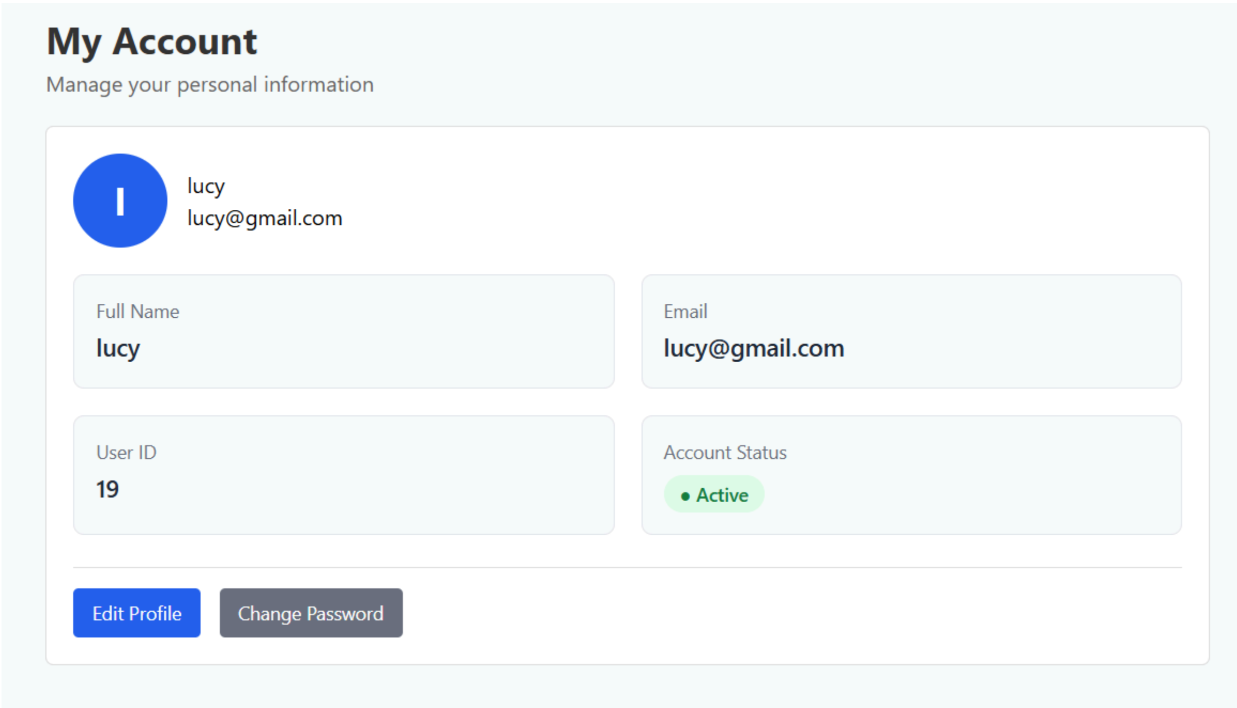
id	active	created_at	email	full_name	password	updated_at	role_id
1	0x01	2025-12-21 14:33:19.495638	Neth.tithyarita@gmail.com	minhminh	$2a$10$bZkXuggjnu681slQE/5.M.ir2UfkowltqYxOizyQleutF6qTR0t3G	2025-12-21 14:33:19.495638	1
2	0x01	2025-12-21 14:37:25.603475	mingming@gmail.com	mingming	$2a$10$YKix9gTmmeWg2rJz0hNfluFP0RdYPrb0as5gP.5UzdBf912K7eaLO	2025-12-21 14:37:25.603475	1
3	0x01	2025-12-21 14:57:43.437605	minhminh@gmail.com	minhminh	$2a$10$3B09LIoj30D/3zQJPIQsa.h.vTRah.xFHRHIQ5FGuI/CY3JdV4cca	2025-12-21 14:57:43.437605	1
4	0x01	2025-12-22 10:50:00.505858	minh@gmail.com	minhminh	$2a$10$aA3kk48zTXjc/fvS2c0WnuUVJk8Q59rq386f2yj40D.ROQw57WX36	2025-12-22 10:50:00.505858	1
5	0x00	2025-12-27 16:17:03.233926	rita@gmail.com	minhminh	$2a$10$mvZzTNj.70nbZxwN3xy1yOb95g2.rr87mvPLnrObcQLePTZjbqD9q	2025-12-27 16:17:03.233926	1
6	0x01	2025-12-29 22:07:17.999442	alex@gmail.com	alex	$2a$10$TvSspRoRQovY1HTVeEB/b.3hTRivuuZMwGowGZT9M3Q80qilDukd2	2025-12-29 22:07:17.999442	1
7	0x01	2025-12-29 22:10:31.612510	alexxx@gmail.com	alex	$2a$10$0AUEd6HruCd3IPZ6tfx/ZO82pCnrzBeVSFXGnRbLF8fNJ0sReksH6	2025-12-29 22:10:31.612510	1
8	0x01	2025-12-29 22:13:50.548355	alexxxx@gmail.com	alex	$2a$10$JKDX8X7Kg6zzwwQ7td/9T.tmu2aH5W7E9XLv8Cl.Utn15.RgsceH2	2025-12-29 22:13:50.548355	1
9	0x01	2025-12-29 22:30:00.577875	mkiii@gmail.com	miiiii	$2a$10$BtXWyMHX0fj61gs.EXLdf.pmy4fpU91jdOFC4LWF.IGXKY/qYgjmS	2025-12-29 22:30:00.577875	1
11	0x01	2026-01-02 15:20:33.125120	lyka@gmail.com	lyka	$2a$10$o9dAZ2DqteZnPr2QQ/ggieU1CTTpWVDTxQCEC1Fcw73ZZZWYfFixO	2026-01-02 15:20:33.125120	1
12	0x01	2026-01-03 09:37:44.162210	lita@gmail.com	lita	$2a$10$GG71CPKFry3pnYLPuo8kSe17QsVkCRylSvWhbiEiSZJO2C4TnJEZW	2026-01-03 09:37:44.162210	1
13	0x01	2026-01-03 22:31:53.658588	librarian@library.com	Default Librarian	$2a$10$eXwluhDO5OA1qVCKU3mXQ.c6s4GjKAcM//jJJBU9udCv.e0ugjo7O	2026-01-03 22:31:53.658588	2
14	0x01	NULL	ysl@gmail.com	ysl	223344	NULL	NULL
15	0x01	2026-01-03 22:53:35.197003	ira@gmail.com	ira	$2a$10$t1//BNUrwu5Zv1JQSrAeZOZ66jUGnx5dXfnZSaL0iy4qkGRvveb5C	2026-01-03 22:53:35.197003	1
16	0x01	2026-01-04 14:30:47.644165	july@gmail.com	july	$2a$10$xLItUyFOTQINoqLq0zZ2Nuy8CKbxu.nVSIYaQFZYIx.gbsbRD7WHG	2026-01-04 14:30:47.644165	1
17	0x01	2026-01-04 14:50:09.722011	june@gmail.com	june	$2a$10$W48SnhhivNgWh7F0WviQkuOoir8cj6mFEb50s9Kn5T95elisUHKcC	2026-01-04 14:50:09.722011	1
18	0x01	2026-01-05 09:46:27.556548	nika@gmail.com	nika	$2a$10$qMZcT4MKvDXK5uD.uyQtnucZfcXCGPXP5V.J/RB28/qTNFZMPw7eq	2026-01-05 09:46:27.556548	1
19	0x01	2026-01-13 11:26:35.571199	lucy@gmail.com	lucy	$2a$10$yWRSO89qtQPC49ruCBhkruGzTXq/wLLT4007yNJNdZa42rxP7pENu	2026-01-13 11:26:35.571199	1

18 rows in set (0.00 sec)

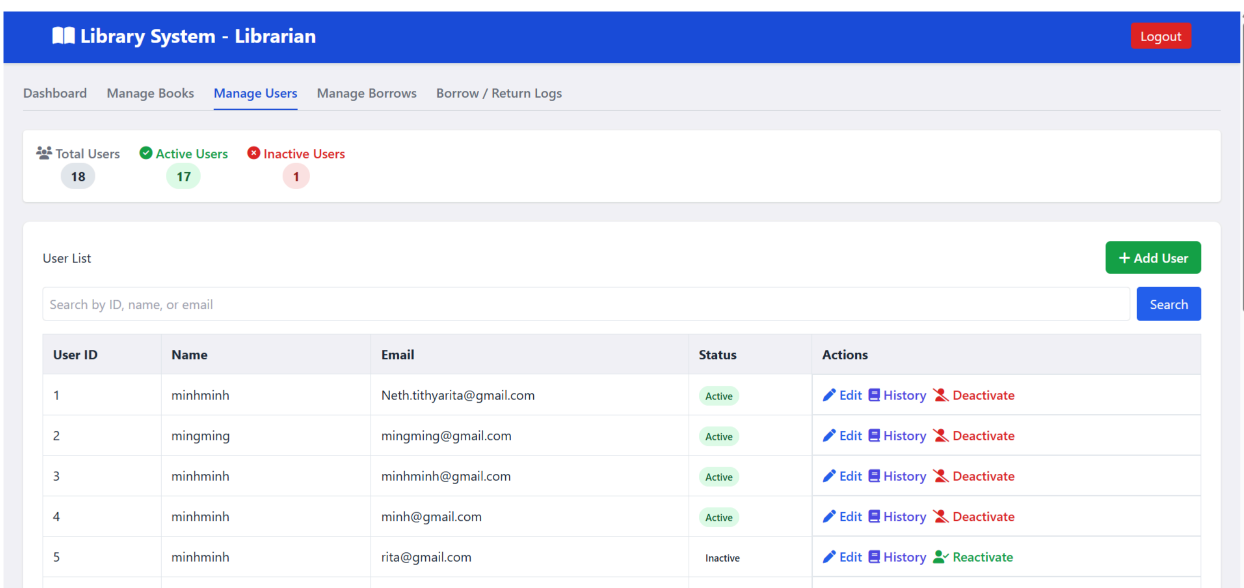
- ★ Next the user can login and access to the library borrow system



❖ Here's the student account in library borrow system



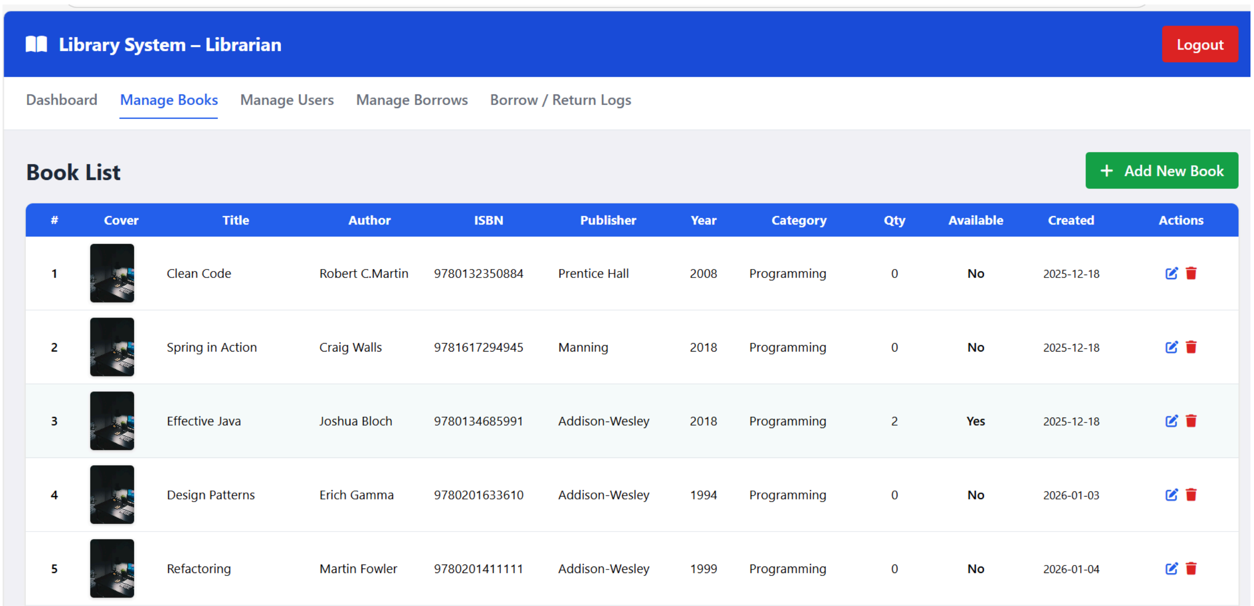
★ Here's librarian dashborad , librarian can manage user(add, edit, view history, search, reactivate and deactivate)



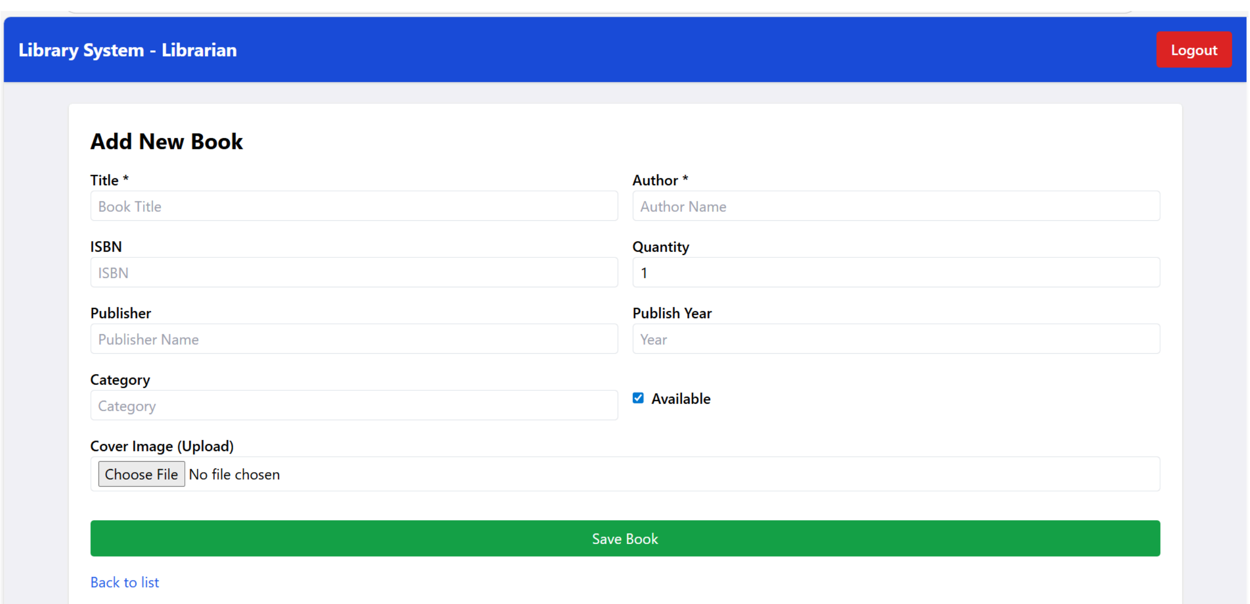
7.2. Book Management

Here's the librarian dashboard, librarians can manage books such as add new books, edit books and delete books.

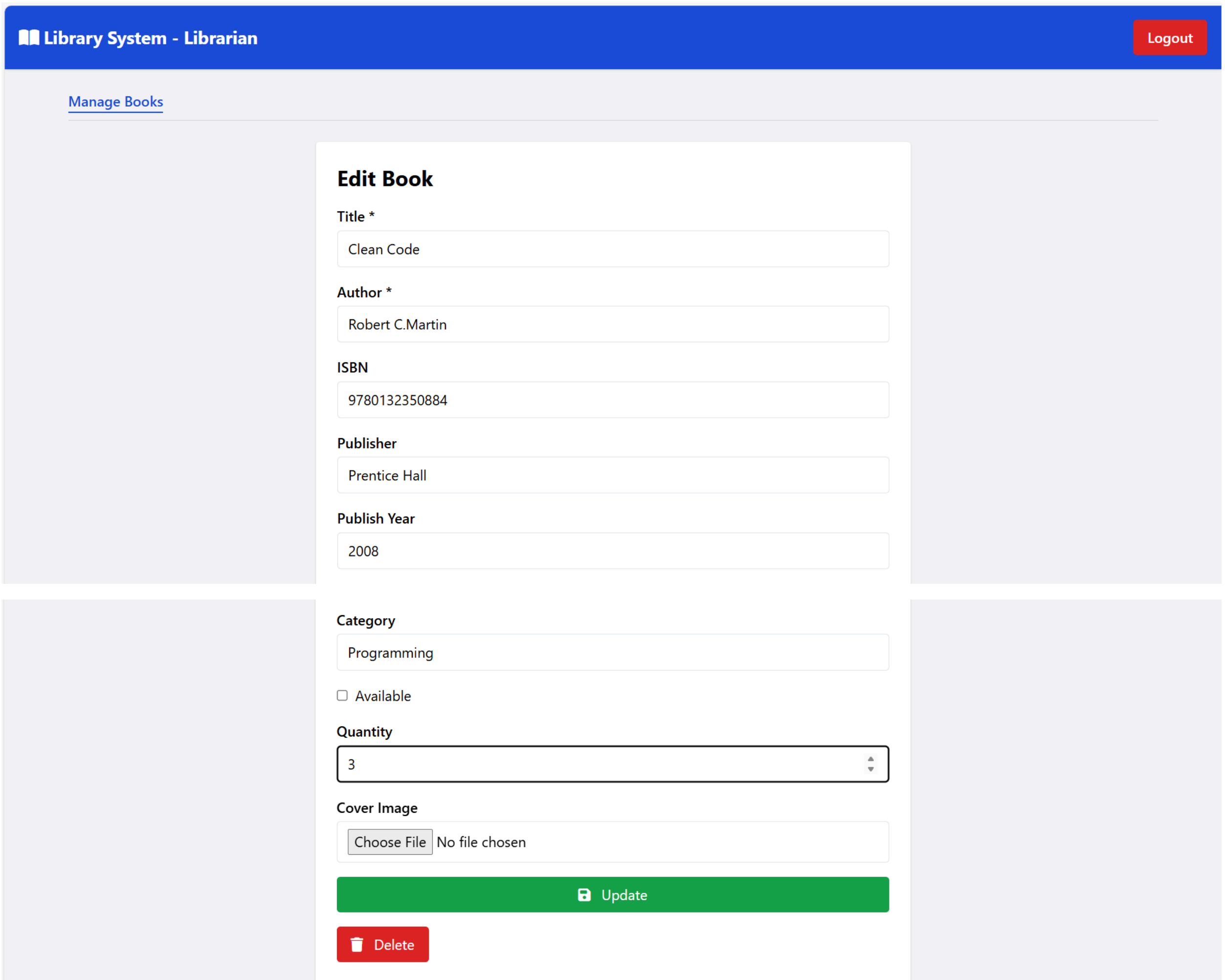
- Manage Books librarians manage all books that are available or unavailable.



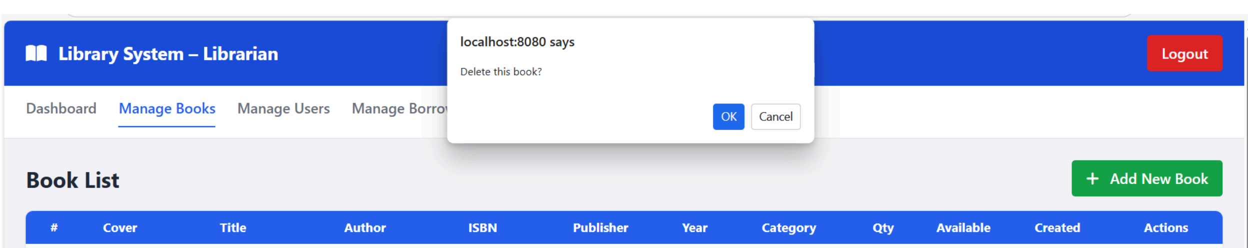
- When the librarian wants to add a new book just click the button Add New Books it has Title, Author, ISBN, Publisher, Publish year, Category,Upload image, Qty, and Available for adding the new book.



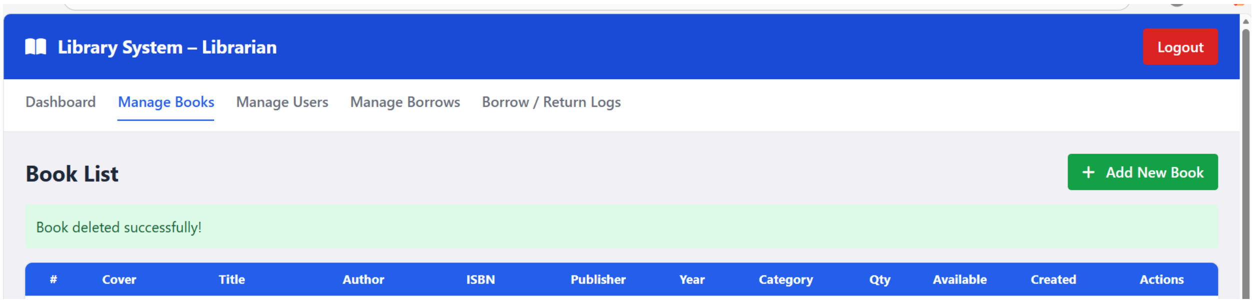
- The librarian can edit the book of some part that they want to edit.



- The librarian also can delete the book.



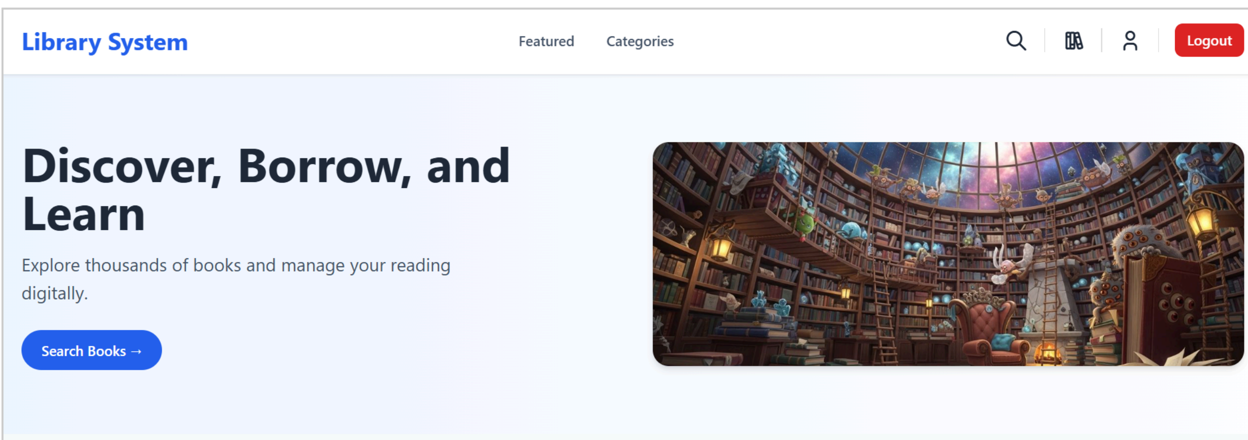
- When click ok it will remove the book that deleted and will give the message like this:



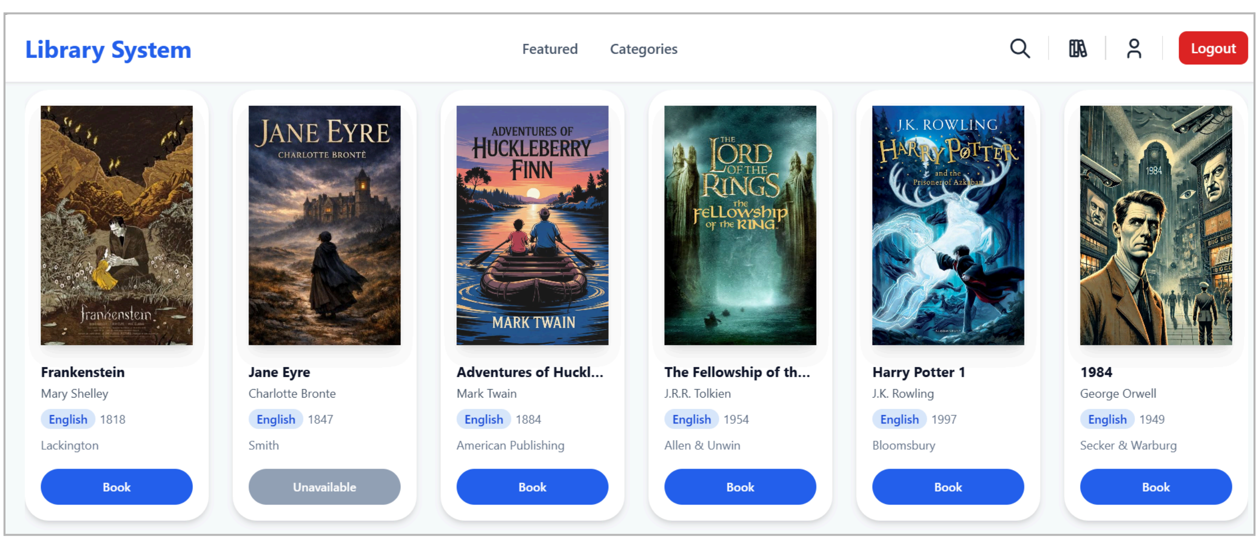
7.3. Browsing and Searching for Books

★ After login as the user they will reach to homepage

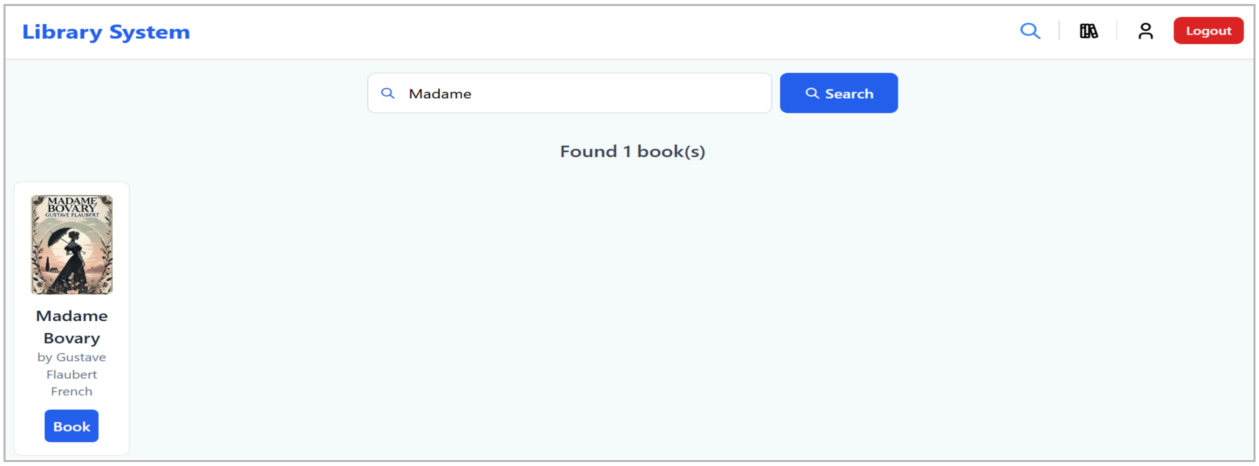
- Homepage



★ The user can scroll down to see all the books on the homepage.



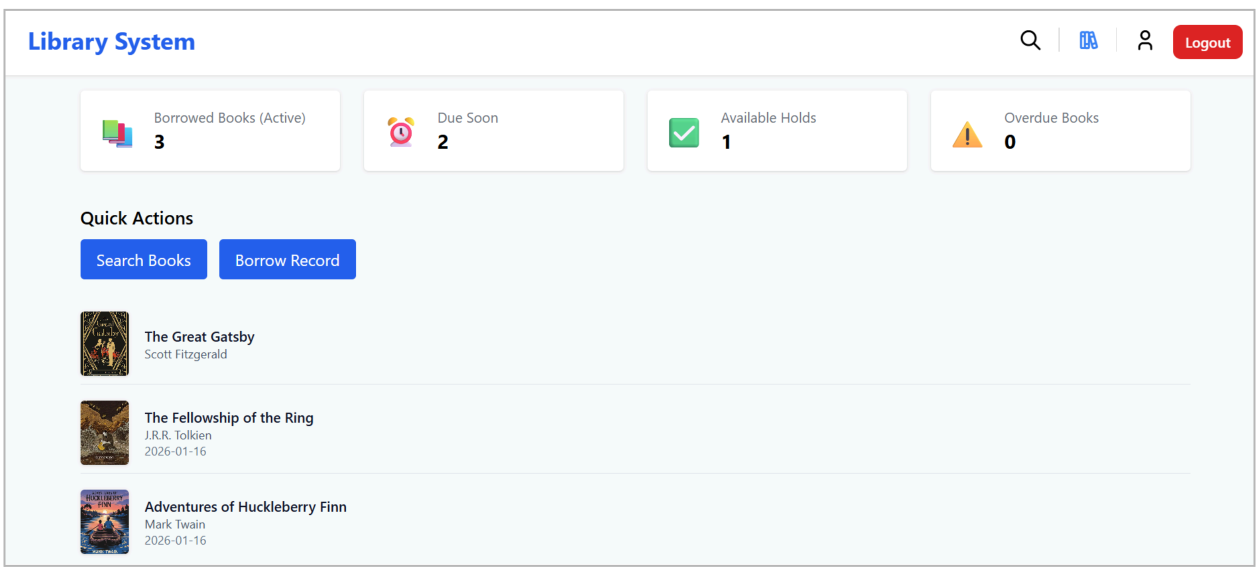
★ Or they can search for the book they want to borrowed



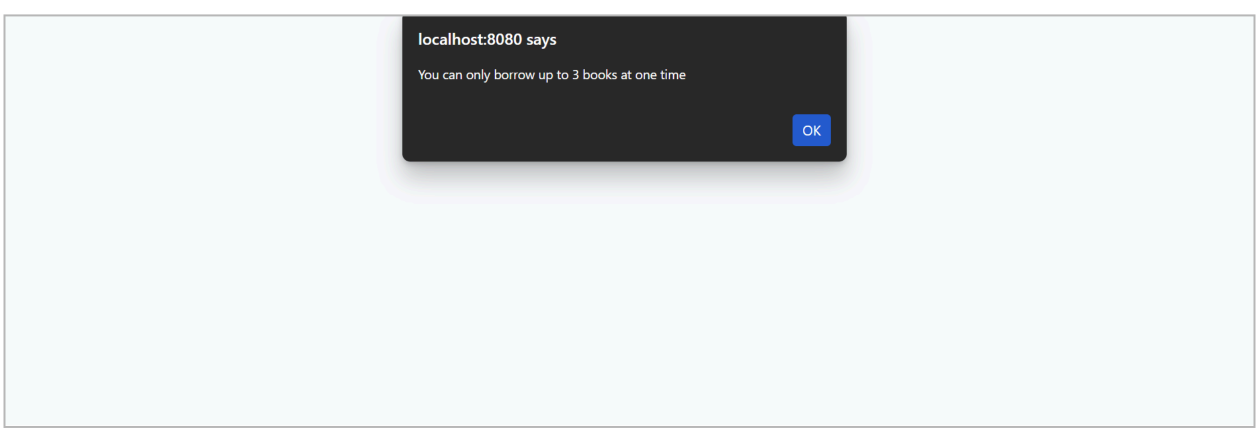
7.4. Borrowing Conditions Check and Record booking

★ The user can see what book are available or not by homepage or search

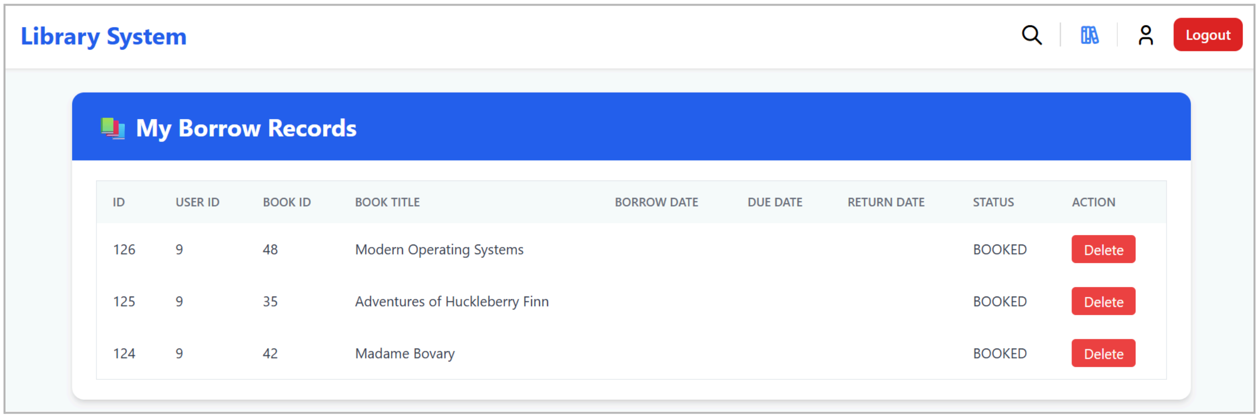
★ So the active user can click on button book then it will go to user dashboard



★ If the user book more than 3



★ User can also see their detail booking record in Borrow Record

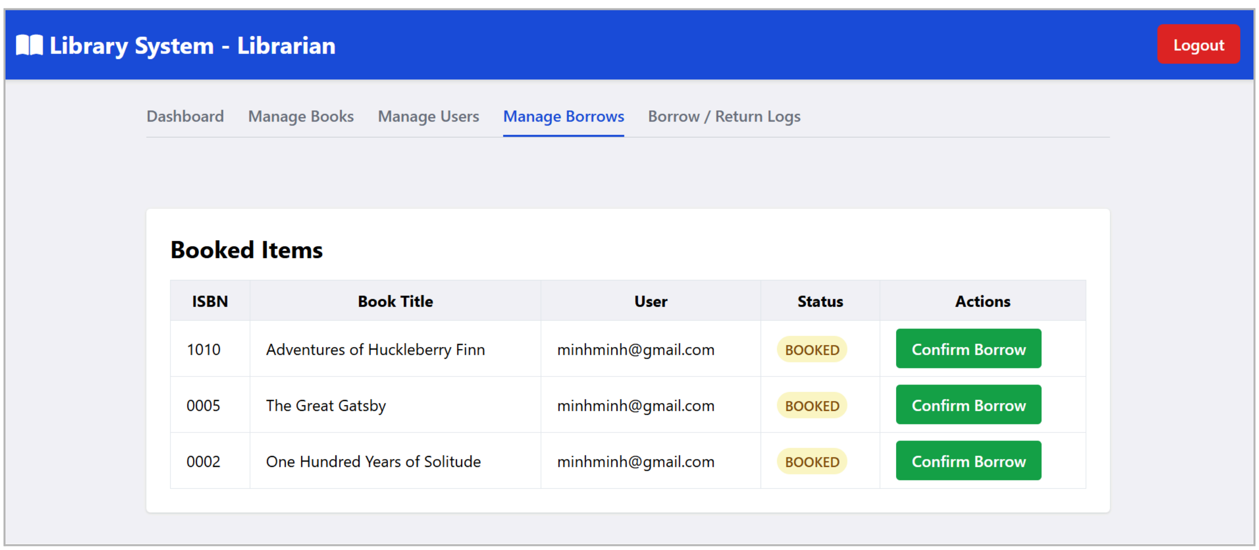


7.5. Confirm booking

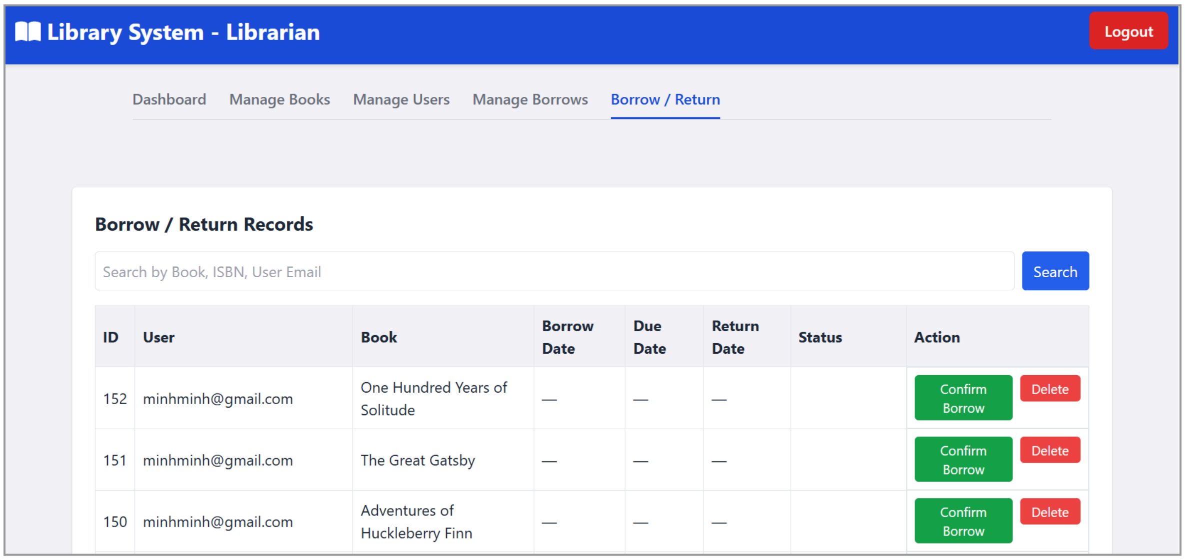
★ So when the user goes to the library to take the book they will check with the librarian about their booking.

★ Librarian will check on

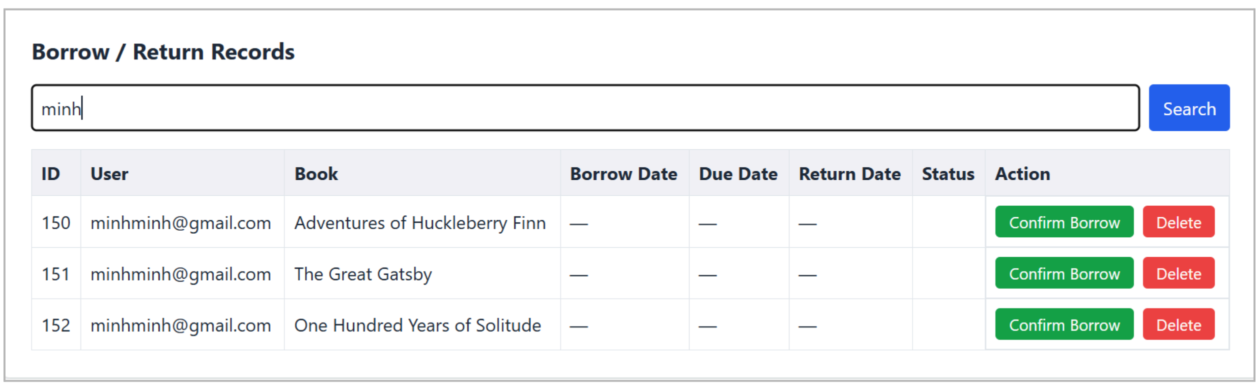
- ○ Manage Borrow (See only all the book that not have been confirm booking)



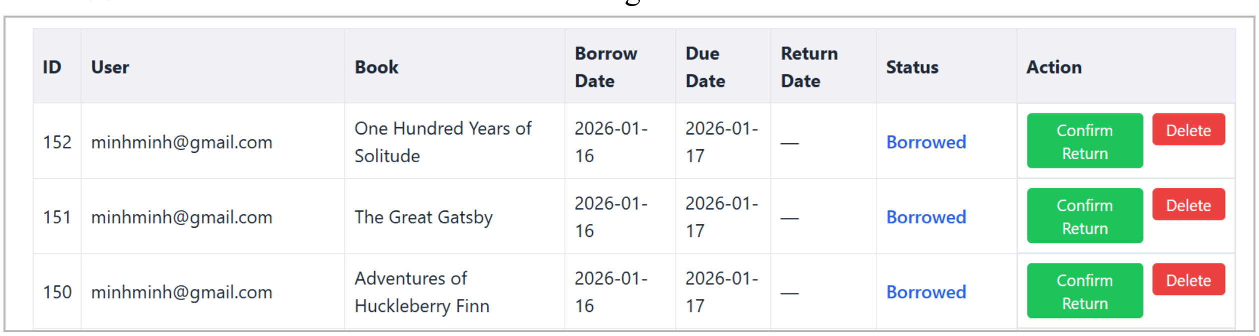
- Borrow/Return (See Borrow confirm, Return confirm and Delete)



- Search borrow



★ When the librarian confirm the booking the date to return start counted



★ The user can also see their borrow record after the librarian confirms. It has been updated the Date, Status and button Delete has disappeared.

My Borrow Records

ID	USER ID	BOOK ID	BOOK TITLE	BORROW DATE	DUE DATE	RETURN DATE	STATUS	ACTION
152	2	27	One Hundred Years of Solitude	2026-01-16	2026-01-17		BORROWED	No actions
151	2	30	The Great Gatsby	2026-01-16	2026-01-17		BORROWED	No actions
150	2	35	Adventures of Huckleberry Finn	2026-01-16	2026-01-17		BORROWED	No actions

7.5. Book Return Process

★ Where users go to the library to return the book they also need to check with librarian.

★ The librarian needs to confirm the return.

- If the user return on time the status will change to RETURN_ON_TIME

126	leanghour12@gmail.com	Modern Operating Systems	2026-01-13	2026-01-27	2026-01-13	Returned on time	Delete
125	leanghour12@gmail.com	Adventures of Huckleberry Finn	2026-01-13	2026-01-27	2026-01-13	Returned on time	Delete
124	leanghour12@gmail.com	Madame Bovary	2026-01-13	2026-01-27	2026-01-29	Returned on time	Delete

My Borrow Records

ID	USER ID	BOOK ID	BOOK TITLE	BORROW DATE	DUE DATE	RETURN DATE	STATUS	ACTION
126	9	48	Modern Operating Systems	2026-01-13	2026-01-27	2026-01-13	RETURNED_ON_TIME	No actions
125	9	35	Adventures of Huckleberry Finn	2026-01-13	2026-01-27	2026-01-13	RETURNED_ON_TIME	No actions
124	9	42	Madame Bovary	2026-01-13	2026-01-27	2026-01-13	RETURNED_ON_TIME	No actions

- If the user return after the due of the date the status will change to RETURN_LATE

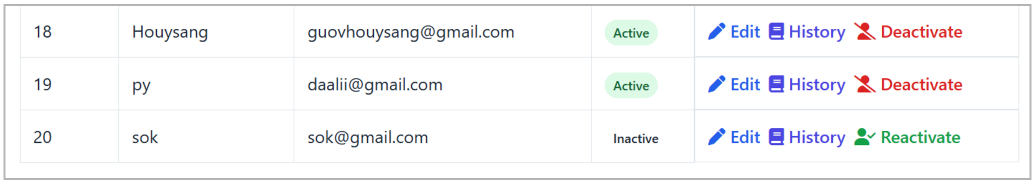
Borrow / Return Records

Search by Book, ISBN, User Email

Search

ID	User	Book	Borrow Date	Due Date	Return Date	Status	Action
143	sok@gmail.com	L'Étranger	2026-01-10	2026-01-11	2026-01-15	Returned late	Delete
142	sok@gmail.com	Madame Bovary	2026-01-10	2026-01-11	2026-01-15	Returned late	Delete
141	sok@gmail.com	Germinal	2026-01-10	2026-01-11	2026-01-15	Returned late	Delete

- If the user return late three time they will be blocked



18	Houysang	guovhouysang@gmail.com	Active	Edit History Deactivate
19	py	daalii@gmail.com	Active	Edit History Deactivate
20	sok	sok@gmail.com	Inactive	Edit History Reactivate

XII. Project Workflow

8.1. Frontend to Backend Flow

This flow explains how a user action in the interface is sent to the Spring Boot backend.

When a student or librarian uses the system:

1. The user performs an action (login, search book, borrow book, return book)
2. The frontend sends an HTTP request (GET, POST, PUT, or DELETE) to a Spring Boot API endpoint
3. The request contains data in **JSON format**
4. The **Controller** in Spring Boot receives the request
5. The Controller sends the request to the **Service layer** for processing
6. The Service layer applies business rules (such as checking availability or user limits)

8.2. Backend to Database Flow

This flow explains how Spring Boot communicates with the MySQL database.

After the Service layer processes the request:

1. The Service sends a request to the **Repository layer**
2. The Repository layer uses **JPA/Hibernate** to communicate with MySQL
3. The database is queried or updated (insert, update, delete, select)
4. MySQL returns the result to the Repository
5. The Repository returns the data to the Service
6. The Service sends the result to the Controller
7. The Controller sends the final response back to the frontend in JSON format

IX. Technical Process

9.1. Tools & Technologies

The Library Borrow and Book Tracking System was developed using modern and reliable tools to ensure efficiency and security.

- Backend Technologies
  - **Java**: Main programming language used to build the system logic.
  - **Spring Boot**: Used to build the backend system and APIs.
  - **Spring MVC**: Helps structure the application using Controller, Service, and Repository layers.
  - **Spring Data JPA/Hibernate**: Used to connect with the MySQL database easily.
  - **Spring Security:** Provides authentication, authorization, and role-based access control for users.

- Frontend Technologies
  - **HTML**: Used to structure web pages.
  - **CSS/Tailwind CSS**: Used to design a clean and responsive user interface.
  - **JavaScript**: Used to handle user actions.
  - **Thymeleaf**: Used to connect frontend pages with Spring Boot backend.

- Database
  - **MySQL**: Used to store all system data (users, books, borrowing records, and roles).

- Development Tools
  - **VS Code**: Code editor for writing and managing the project.
  - **Grandle**: Used to manage libraries and run the project.
  - **MySQL Workbench**: Used to manage the database.
  - **Git**: Used to track code changes.
  - **Web Browser**: Used to test the system.

9.2. Spring Boot Workflow

The Spring Boot Workflow describes how a user request is processed inside the system.

1. The user logs in, searches a book, borrows, and returns the book.
2. The frontend sends a request to the backend.
3. The **Controller** receives the request.

4. The Controller sends the request to the **Service**.
5. The Service checks system rules:
    - Book availability
    - Borrow limit (maximum 3 books)
    - User status (active or blacklisted)

6. The Service sends data to the **Repository**.
7. The Repository communicates with the **MySQL database**.
8. The database sends the result back.
9. The result is returned to the frontend.
10. The user sees the final result on the screen.

X. Challenge

10.1. Strengths

Strengths of the Library Borrow and Book Tracking System:

- **Efficient Book Management**: The system allows librarians to manage books,manage users ,manage borrow and borrow/return log, and monitor availability easily, reducing manual work and errors.
- **Borrowing Limit Control**: By restricting users to borrow a maximum of **three books at a time**, the system ensures fair access to library resources for all users.
- **Easy Monitoring of Borrowed Books**: Librarians can quickly identify which books are borrowed, by whom, when they were borrowed and when they were returned.
- **User Authentication and Security**: The system supports user login and role management (librarian/user), which protects sensitive data and prevents unauthorized access.
- **Scalable System Design**: Built using **Spring Boot and MVC architecture**, the system can be extended with more features in the future.

10.2. Weakness

Weaknesses of the Library Borrowing and Book Tracking System:

- **Limited Borrowing Features**: The current system only supports basic borrowing and returning, without advanced features such as reservations or renewals.
- **Dependency on Internet and Server Availability**: If the server or network is unavailable, users and librarians cannot access the system.
- **No Notification System**: Users do not receive reminders for due dates or overdue books via email or SMS.

XI. Conclusion

The Library Borrowing and Book Tracking System was designed to improve the management of library operations by automating the borrowing and tracking of books. The system helps librarians and users manage book records efficiently while reducing the need for manual paperwork. By maintaining all borrowing information in a centralized database, the system ensures accuracy and reliability of data.

One of the key features of the system is the limitation that allows a user to borrow a maximum of three books at a time. This rule promotes fair usage of library resources and prevents misuse. The system also provides secure user authentication, ensuring that only authorized users can access specific features.

Overall, the project successfully demonstrates how a web-based system can enhance efficiency, organization, and control in a library environment. With future enhancements such as fine management, notifications, the system can become a more advanced and complete library management solution.